ch11 Uniform Circular Motion

Q3

Please stick the barcode label here.

*3.

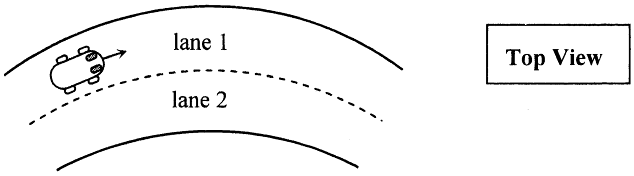


Figure 3.1

Figure 3.1 shows the top view of a horizontal road with two circular lanes. A car of mass 1200 kg moves with constant speed in lane 1 of radius 45 m.

(a) (i) Name the force that provides the centripetal force for the car. If the maximum value of this force is 8000 N, calculate the highest speed of the car such that it can keep in lane 1. (3 marks)

(ii) Suppose the car takes lane 2 instead of lane 1 and the maximum value of the force providing the centripetal force is still 8000 N. Would the car's highest speed in lane 2 be smaller than, larger than or the same as that found in (a)(i) ? Explain. (2 marks)

(b) Explain why the chance of skidding would increase if there are oil patches on the road surface in Figure 3.1. (2 marks)

Answers written in the margins will not be marked.

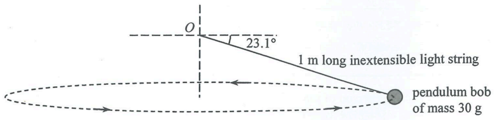
Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Q4

*4. (a)

Figure 4.1



A pendulum bob of mass 30 g is tied to a fixed point O by a 1 m long inextensible light string. It is swirled to describe a horizontal circle uniformly at an angular velocity of 5.0 rad s^{-1} as shown in Figure 4.1. Neglect air resistance. ($g = 9.81$ m s^{-2})

(i) What is the bob's rotation rate (in revolutions per second) ? (1 mark)

(ii) Indicate on Figure 4.1 the centripetal force F_C required for the motion of the bob. Find F_C. (3 marks)

(iii) Explain whether the magnitude of the tension in the string is greater than, equal to or smaller than the centripetal force F_C found in (a)(ii). (2 marks)

Answers written in the margins will not be marked.

Please stick the barcode label he

(b) The moon is orbiting around the Earth uniformly in a circular path under the influence of the Earth's gravitational attraction.

(i) Explain why the speed of the moon remains unchanged although it is acted upon by gravitational force. (2 marks)

(ii) A student claimed that as the moon is much less massive than the Earth, it exerts negligible force on the Earth. Comment on the student's claim. (2 marks)

Answers written in the margins will not be marked.

Q4

*4. (a) Figure 4.1(a) shows a hollow cylindrical drum rotating uniformly about its central vertical axis. With a high enough rotation speed, the block shown in the figure appears 'pinned' to the wall of the drum without falling.

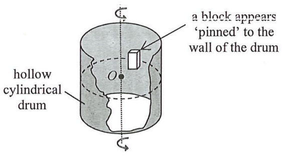


Figure 4.1 (a)

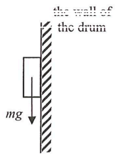


Figure 4.1 (b)

(i) Besides the block's own weight mg, there are two more forces acting on it. Indicate and label these two forces on Figure 4.1(b). (2 marks)

(ii) Which force provides the centripetal force for the block ? (1 mark)

....

....

(b) A washing machine has a hollow cylindrical drum with numerous small holes that allow water droplets to escape. In 'spin' mode, the drum's angular speed ω is 78.5 rad s^{-1}.

(i) What is the rotation speed of the drum in revolutions per minute ? (2 marks)

....

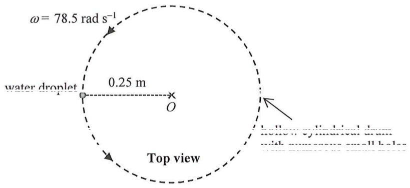
....

....

....

The cylindrical drum of the washing machine is of radius 0.25 m. Figure 4.2 shows the top view of the drum spinning anticlockwise about its central vertical axis through O.

Figure 4.2



(ii) Estimate the centripetal force acting on the water droplet shown in Figure 4.2. The mass of the droplet is 2×10^{-5} kg. (2 marks)

....

....

....

....

....

Suppose that the water droplet is about to escape via a small hole on the wall of the drum,

(iii) estimate the speed of the water droplet. (2 marks)

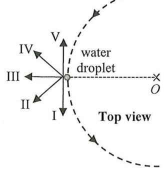
....

....

....

....

(iv) along which direction (I – V) would the droplet travel just as it escapes from the drum ? (1 mark)



....

....

Q4

Please stick the barcode label here.

*4. A sphere of radius 0.50 m rotates about a vertical axis ACB with a constant angular speed of 6 rad s^{-1}, with C being the sphere's centre. A small plasticine P of mass 0.020 kg is stuck on the surface of the sphere such that it rotates on a horizontal plane containing C as shown in Figure 4.1(a).

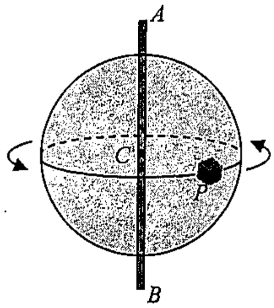


Figure 4.1(a)

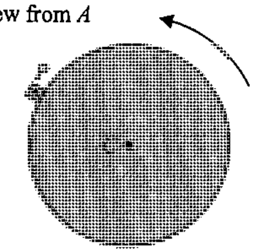


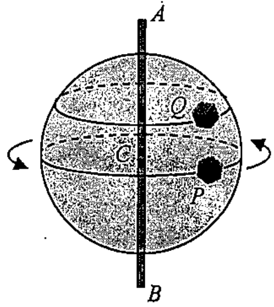
Figure 4.1(b)

(a) (i) Find the magnitude of the centripetal force acting on P. (2 marks)

(ii) The sphere's angular speed is increasing. If P finally leaves the sphere at the position shown in Figure 4.1(b) (Top view from A). Indicate on Figure 4.1(b) the direction of motion of P when it **just** leaves the sphere. (1 mark)

(b) Another plasticine Q is stuck on the sphere's surface as shown in Figure 4.2.

Figure 4.2



(i) State whether P and Q have the same angular speed. (1 mark)

(ii) Explain whether P and Q have the same magnitude of centripetal acceleration. (2 marks)

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.

Answers written in the margins will not be marked.